

Logic-based Benders decomposition for doctor-patient matching and scheduling considering chronic patients' online consultation time preference

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Problem description

Digital healthcare

Online consultation platform for patients with **chronic diseases**.

- Patients have preferences for consultation times and doctors
- Doctors have preferences for diseases and varying appointment lengths for each disease

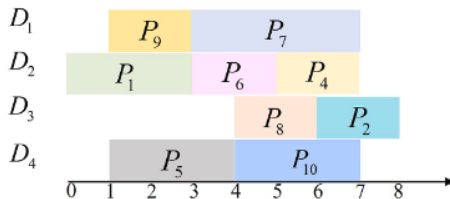


Figure: Example schedule

Problem description

Multiple objectives:

- ① Maximise patient satisfaction:
 - Satisfaction with the consultation time
 - Satisfaction with the doctor allocation
- ② Maximise doctor satisfaction
 - Satisfaction with the disease treated
- ③ Maximise number of successful matches

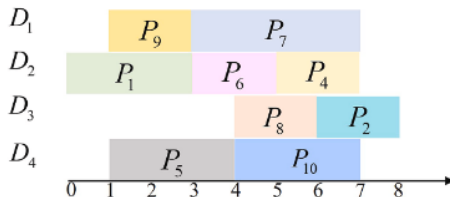


Figure: Example schedule

Model formulation

Sets:

- I Patients
- J Doctors
- K Disease types
- T Time periods
- $I_k \subset I$ Patients who have disease $k \in K$

Variables:

- $x_{i,j,t}$: 1 if patient $i \in I$ **is in an appointment with** doctor $j \in J$ in time $t \in T$, 0 otherwise
- $y_{i,j,t}$: 1 if patient $i \in I$ **starts** an appointment with doctor $j \in J$ in time $t \in T$, 0 otherwise

Model formulation

Objective and constraints that use the x_{ijt} variable:

Objective: Patient preferences

$$\sum_{i \in I} \sum_{j \in J} \sum_{t \in T} \text{patientTimeScore}_{it} x_{ijt} + \sum_{i \in I} \sum_{j \in J} \sum_{t \in T} \text{patientDoctorScore}_{ij} y_{ijt}$$

Constraints:

- Patients are not overbooked:

$$\sum_{j \in J} x_{ijt} \leq 1 \quad \forall i \in I, \forall t \in T$$

- Link x and y . If an appointment starts then it lasts the duration:

$$\sum_{t' \in [t, t + \text{treat}_{j,k} - 1]} x_{i,j,t'} \geq \text{treat}_{j,k} y_{ijt} \quad \forall k \in K, \forall i \in I_k, \forall j \in J, \forall t \in T$$

Solution technique

Multiple objectives: ε -constraint method

- Patient satisfaction is set as the primary objective (multidimensional).
- Doctor satisfaction and the number of matches are *transformed into constraints*, with lower bounds $\varepsilon_1 \in \mathbb{R}$ and $\varepsilon_2 \in \mathbb{Z}$
- **Pareto optimal solutions:** Trace out the Pareto frontier by systematically varying the values of ε_1 and ε_2

Logic-based Benders decomposition

- **Master problem (matching):** Determine patient-doctor matches, maximising patient satisfaction with consultation time.
- **Sub-problem (scheduling):** Determine consultation start times, maximising patient satisfaction with doctor and service duration.
- **Logic-based cutting inequality:** Prevent cases where a doctor-patient match is included in one iteration but excluded later.

Ideas for improvement

- **Simplification:** Use a variable only for the *start* of consultation.
 - Avoids tracking ongoing interaction in each period.
 - Fixed consultation durations handle this automatically.
- **Extension:** Reversing iteration order in generating Pareto frontier.
 - First iterate on number of matches (discrete ε_2).
 - Then doctor satisfaction (continuous ε_1).
 - Potentially yields more structured approximations.

Our model formulation

Sets:

- I Patients
- J Doctors
- K Disease types
- T Time periods
- $I_k \subset I$ patients which have disease $k \in K$

Data:

- Service times of qualified doctors ($\text{treat}_{j,k}$)
- Doctor ranking (over diseases treated)
- Patient ranking (over feasible times and doctor allocation)
- Time windows for consultations (doctor and patient availability)

Decision variable:

- $y_{i,j,t} \in \{0, 1\}$ patient-doctor match (i, j) begins in time period t

Constraints:

- Patients are seen at most once:

$$\sum_{j \in J} \sum_{t \in T} y_{i,j,t} \leq 1 \quad \forall i \in I$$

- Doctors only treat patients for diseases they are qualified to treat:

$$y_{i,j,t} \leq \text{qualified}_{j,k} \forall k \in K, i \in I^k, j \in J, t \in T$$

Our model formulation

Constraints:

- Doctors only treat one patient at a time:

$$\sum_{k \in K} \sum_{i \in I^k} \sum_{t' \in [\text{minService}, t]} y_{i,j,t'} \leq 1 \quad \forall j \in J, t \in T$$

- Feasible time periods:

$$\begin{aligned} \text{doctorTime}_{j,t'} \cdot \text{patientTime}_{i,t'} &\geq y_{i,j,t} \\ \forall k \in K, i \in I^k, j \in J, t \in T, t' \in [t, t_{\text{maxService}}] \end{aligned}$$

- $\text{minService} = \max\{0, t - \text{treat}_{j,k} + 1\}$ is the first service period
- $\text{maxService} = \min\{t + \text{treat}_{j,k} + 1, |T|\}$ is the last service period.

Our model formulation

We want to change the Patient satisfaction objective function.

Their objective function

$$\sum_{i \in I} \sum_{j \in J} \sum_{t \in T} \text{patientTimeScore}_{it} x_{ijt} + \sum_{i \in I} \sum_{j \in J} \sum_{t \in T} \text{patientDoctorScore}_{ij} y_{ijt}$$

Our objective function

$$\sum_{i \in I} \sum_{j \in J} \sum_{t \in T} (\text{patientTimeScore}_{it} + \text{patientDoctorScore}_{ij}) y_{ijt}$$

We also want to change the calculations of $\text{patientTimeScore}_{it}$ and $\text{patientDoctorScore}_{ij}$.

We have replicated the data generation algorithms and will benchmark our algorithm against the same experimental setup.

- **Data:** The authors generate synthetic data
 - Partly arbitrary (parameters are randomly, uniformly distributed).
 - Informed by digital healthcare platform statistics.
- **Simulation:** Instance sizes vary by number of patients and doctors.

Thank you

Reference: Jiang et al. (2025). *Logic-based Benders decomposition for doctor-patient matching and scheduling considering chronic patients' online consultation time preference*. Computers & Operations Research 183: 107207. <https://doi.org/10.1016/j.cor.2025.107207>